AI-Based Smart Hostel Complaint Management System

Project Overview

The ****AI-Based Smart Hostel Complaint Management System**** is a comprehensive, modern application designed to streamline the process of reporting, tracking, and resolving issues within a hostel environment. The platform features an intelligent backend and a cross-platform mobile application, providing dedicated interfaces for different user roles such as Students and Wardens.

By leveraging Artificial Intelligence (Groq API), the system optimizes complaint categorization, predicts resolution times, and ensures smart resource allocation, creating a hassle-free experience for hostel administration and residents alike.

🛠️ Technology Stack

The application employs a robust, scalable, and modern technology stack:

Frontend (Mobile App)

- ****Framework:**** Flutter / Dart (Cross-platform for Android & iOS)
- ****State Management:**** Provider
- ****Routing:**** Go Router
- ****Networking:**** Dio & HTTP Client
- ****Local Storage:**** Flutter Secure Storage
- ****UI/UX:**** Google Fonts, Fl_chart (for Analytics), Lottie & Animations, Shimmer (for loading states)

Backend (RESTful API)

- ****Runtime:**** Node.js
- ****Framework:**** Express.js
- ****Architecture:**** MVC (Model-View-Controller)
- ****AI Integration:**** Groq SDK (for AI-driven insights and categorization)

- **Hosting / Deployment:** Vercel

Database & Cloud Services

- **Database:** MongoDB (via Mongoose ODM)

- **Authentication & Notifications:** Firebase Admin SDK, Google Auth Library, JWT (JSON Web Tokens)

- **Media Storage:** Cloudinary (for securely storing complaint images and attachments)

👥 User Roles & Features

1. Student Module

- **Authentication:** Secure Google Sign-In and email authentication.

- **Complaint Registration:** Easy submission of complaints with attached images, descriptions, and category tags.

- **Real-Time Tracking:** Live status updates (e.g., Pending, In Progress, Resolved).

- **Notifications:** Push notifications via Firebase Cloud Messaging (FCM) when the status of a complaint changes.

- **Feedback System:** Provide feedback and ratings on resolved complaints to maintain quality of service.

2. Warden / Administration Module

- **Dashboard & Analytics:** Visual charts (Fl_chart) showing complaint trends, pending issues, and resolution efficiency.

- **Complaint Management:** Ability to view, assign, and update the status of complaints.

- **AI-Powered Insights:** AI helps in categorizing complaints and analyzing historical data for preventive maintenance.

- **Staff Management:** Assignment of complaints to specific maintenance staff or departments (e.g., Electrical, Plumbing, Cleaning).

📐 Core Architecture & Data Models

The system is built on a NoSQL document database. Key data models include:

- **User:** Manages roles (Student, Warden), personal details, and authentication tokens.

- **Complaint:** Stores the core issue, including title, description, category, attached images (Cloudinary URLs), assigned staff, and current status.

- **ComplaintHistory:** An audit trail of every status change and action taken on a complaint, ensuring complete transparency.

- **Department & MaintenanceStaff:** Organizes the workforce resolving the issues.

- **Feedback:** Captures student satisfaction post-resolution.

- **Notification:** Manages alerts triggered during the complaint lifecycle.

🔒 Security & Authentication

Security is a top priority for this system:

1. **JWT Auth:** JSON Web Tokens are used for secure session management across all API endpoints.

2. **Role-Based Access Control (RBAC):** Distinct privileges ensuring Students cannot access Warden-level dashboards.

3. **Data Protection:** Passwords and sensitive data are secured using industry-standard hashing.

4. **Rate Limiting & Helmet:** Backend is secured against brute-force attacks and common web vulnerabilities using `express-rate-limit` and `helmet`.

🚀 Deployment Workflow

- **Backend:** The Node.js/Express server is hosted on **Vercel**, enabling seamless CI/CD, automatic scaling, and fast response times.

- **Database:** Hosted on **MongoDB Atlas**, providing a fully managed cloud database solution.

- **Frontend:** Built with **Flutter**, allowing compilation to native ARM code for both Android (APK/AAB) and iOS (IPA).

Conclusion

This AI-Based Smart Hostel Complaint Management System represents a significant technological leap in institutional administration. By bridging Flutter's dynamic UI, Node.js's high-performance backend, MongoDB's flexible data storage, and Groq's cutting-edge AI capabilities, it offers a seamless, scalable, and highly efficient solution for modern hostel environments.